| VII SEMESTER (PROFESSIONAL ELECTIVE-V) | L | T | P | C |
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| **20AM7E09 - SEMANTIC WEB** | | | | |

**COURSE OBJECTIVES:**
1. To learn Web Intelligence
2. To learn Knowledge Representation for the Semantic Web
3. To learn Ontology Engineering
4. To learn Semantic Web Applications, Services and Technology
5. To learn Social Network Analysis and semantic web

**COURSE OUTCOMES:**
**After the completion of the course, student will be able to**
**CO1:** Demonstrate social network analysis and measures.
**CO2:** Analyze random graph models and navigate social networks data
**CO3:** Apply the network topology and Visualization tools.
**CO4:** Analyze the experiment with small world models and clustering models.
**CO5:** Compare the application driven virtual communities from social network Structure.

**UNIT -I:**
Web Intelligence: Thinking and Intelligent Web Applications, The Information Age ,The World Wide Web, Limitations of Today's Web, The Next Generation Web, Machine Intelligence, Artificial Intelligence, Ontology, Inference engines, Software Agents, Berners-Lee www, Semantic Road Map, Logic on the semantic Web.

**UNIT -II:**
Knowledge Representation for the Semantic Web: Ontologies and their role in the semantic web, Ontologies Languages for the Semantic Web –Resource Description Framework(RDF) / RDF Schema, Ontology Web Language(OWL), UML, XML/XML Schema.

**UNIT -III:**
Ontology Engineering: Ontology Engineering, Constructing Ontology, Ontology Development Tools, Ontology Methods, Ontology Sharing and Merging, Ontology Libraries and Ontology Mapping, Logic, Rule and Inference Engines.

**UNIT -IV:**
Semantic Web Applications, Services and Technology: Semantic Web applications and services, Semantic Search, e-learning, Semantic Bioinformatics, Knowledge Base ,XML Based Web Services, Creating an OWL-S Ontology for Web Services, Semantic Search Technology, Web Search Agents and Semantic Methods,

**UNIT -V:**
Social Network Analysis and semantic web: What is social Networks analysis, development of the social networks analysis, Electronic Sources for Network Analysis – Electronic Discussion networks, Blogs and Online Communities, Web Based Networks, Building Semantic Web Applications with social network features.

**Text Books:**
1. Thinking on the Web – Berners Lee, Godel and Turing, Wiley inter science, 2008.
2. Social Networks and the Semantic Web, Peter Mika, Springer, 2007.

**Reference Books:**

1. Semantic Web Technologies, Trends and Research in Ontology Based Systems, J. Davies, R. Studer,
2. P. Warren, John Wiley & Sons.
3. Semantic Web and Semantic Web Services -Liyang Lu Chapman and Hall/CRC Publishers,(Taylor & Francis Group)
4. Information sharing on the semantic Web – Heiner Stucken schmidt; Frank Van Harmelen, Springer Publications.
5. Programming the Semantic Web, T. Segaran, C. Evans, J. Taylor, O'Reilly, SPD.